P2P-VIP-01 — expert review pack

Vendor invoice processing · development grid, instrument version v1 · n=3 per condition · generated from 6 runs

Runs summarised

baseline: 20260828-175146-baseline-P2P-VIP-01-fd37, 20260830-022032-baseline-P2P-VIP-01-b43c, 20260830-022117-baseline-P2P-VIP-01-7fc2
council: 20260829-051950-council-P2P-VIP-01-13b0, 20260830-022209-council-P2P-VIP-01-4c1d, 20260830-052308-council-P2P-VIP-01-2023

Commits: 24be37cf, 561d5151, 9b06cd86 · model as resolved by probe: claude-sonnet-5

Representative run selected by median per-condition depth; where two runs tie at the median, the earlier timestamp is used. Ties occurred in five of six cells owing to the coarse depth quantum.

Representative runs: baseline **20260830-022032-baseline-P2P-VIP-01-b43c** (depth 0.500, tie broken by timestamp); council **20260830-022209-council-P2P-VIP-01-4c1d** (depth 1.000, tie broken by timestamp).

Measured outcomes, n=3 per condition

Measure	Baseline	Council
Rule conformance (mean)	1.00	1.00
Substantive conformance (mean)	1.00	1.00
Vacuous passes (total across 3 runs)	0	0
Transformation depth, mean (range)	0.458 (0.375–0.500)	0.958 (0.875–1.000)
Actor reassignments (representative run)	4 (varies across runs: [3, 3, 4])	4 (varies across runs: [4, 4, 6])
Output tokens incl. reasoning, mean (range)	5,432 (5,147–5,891)	83,077 (77,835–88,437)

Reading

Depth separates on this workflow: the baseline range (0.375 to 0.500) and the council range (0.875 to 1.000) are disjoint at n=3. This is the only workflow in the development grid where the ranges do not overlap.

Rule conformance is 1.00 in every cell. It is reported as the validity floor — evidence that no redesign in either condition breached a stated requirement — not as a discriminating measure.

Zero-elimination note. Across both instrument versions, 0 of 176 dispositions in 22 completed runs were recorded eliminated and 0 absorbed. Round-one transcripts show the maximalist role explicitly reasoning that no elimination is defensible on this corpus — reasoned declination, not suppression. The labelling-fidelity hypothesis behind instrument v2 is retracted with evidence.

Design comparison — representative runs

Baseline step	Baseline	Council
VIP-01 Invoice receipt and intake AP clerk	**merged** → TVIP-01 Intelligent invoice capture and validation *(AP clerk (with AI/OCR capture engine))*	**merged** → TS-01 Invoice intake and automated extraction *(Intake/extraction agent, with AP clerk exception review)*
VIP-02 Data extraction and validation AP clerk (with OCR/validation engine)	**merged** → TVIP-01 Intelligent invoice capture and validation *(AP clerk (with AI/OCR capture engine))*	**merged** → TS-01 Invoice intake and automated extraction *(Intake/extraction agent, with AP clerk exception review)*
VIP-03 Purchase order matching AP analyst (with ERP matching engine)	**restructured** → TVIP-02 Automated three-way matching *(AP analyst (with ERP/AI matching engine))*	**restructured** → TS-02 Automated three-way matching *(AP analyst, with automated matching engine)*
VIP-04 Exception handling and resolution AP analyst	**restructured** → TVIP-03 Exception investigation and resolution *(AP analyst)*	**restructured** → TS-03 Exception investigation and resolution *(AP analyst)*
VIP-05 Approval routing and authorisation Approver hierarchy	**retained** → TVIP-04 Approval routing and authorisation *(Approver hierarchy)*	**restructured** → TS-04 Risk-tiered approval and authorisation *(Approver hierarchy: policy-delegation engine for the low-risk tier, human approver for the standard/exception tier)*
VIP-06 GL coding and posting AP accountant	**retained** → TVIP-05 GL coding and posting *(AP accountant (posting specialist))*	**restructured** → TS-05 GL coding and posting *(AP accountant)*

Baseline step	Baseline	Council
VIP-07 Payment scheduling and execution Treasury/AP payments officer	**retained** → TVIP-06 Payment scheduling and execution *(Treasury/AP payments officer)*	**restructured** → TS-06 Payment scheduling and execution *(Treasury/AP payments officer)*
VIP-08 Reconciliation and period close AP accountant	**retained** → TVIP-07 Reconciliation and period close *(AP accountant)*	**restructured** → TS-07 Reconciliation and period close *(AP accountant)*

Council-only design elements

10 controls present in the representative council target state and absent from **all three** baseline draws. Absence is verified mechanically against each baseline target state at generation time; a claim that appears in any baseline draw is not emitted.

Control	Where	Argued by	KB
Validation exception queue at capture	TS-01 output	agentic_architect, hitl_designer, finance_domain	—
Variance packet assembled before handoff	TS-02 output, TS-03 input	agentic_architect, finance_domain	—
Structured attributable resolution record	TS-03 output	hitl_designer, finance_domain, accounting_standards	KB-02, KB-05
Risk-tiered approval replacing uniform routing	TS-04 step name	hitl_designer, finance_domain, pragmatist, emerging_economy	KB-02
Auto-authorisation rule-fired log	TS-04 output	finance_domain, agentic_architect	KB-02
Mandatory sampling audit of the auto-approved population	TS-04 output	pragmatist, agentic_architect, hitl_designer	—
Governed delegation-of-authority policy engine for the low-risk tier	TS-04 actor	hitl_designer, emerging_economy, agentic_architect	—
Offline-cached delegation policy with named fallback approver	TS-04 input	emerging_economy, hitl_designer	—
Matching kept as a discrete auditable step boundary	TS-02 description	finance_domain, pragmatist, accounting_standards	KB-09
Discrete subledger lock preserved under continuous reconciliation	TS-07 output	finance_domain, accounting_standards, agentic_architect	KB-03

Strongest rationale per control

Validation exception queue at capture

> *“Anything below the confidence threshold is not silently passed forward — it is routed to an AP-clerk exception queue with the specific low-confidence fields flagged, not the whole document.” — agentic_architect, round 1*

Variance packet assembled before handoff

> *“when a match falls outside tolerance, the same agent assembles the variance packet (PO, GRN, invoice, computed variance, historical vendor pattern) before handing off, rather than dropping a bare "exception" flag” — agentic_architect, round 1*

Structured attributable resolution record

> *“resolution rationale (VIP-04) and reconciliation sign-off (VIP-08) must be captured as structured, attributable records, not narrative comments buried in a ticket system” — hitl_designer, round 1*

Risk-tiered approval replacing uniform routing

> *“Today's step implies uniform hierarchical routing for everything matched. I want…” — hitl_designer, round 1*

Auto-authorisation rule-fired log

> *“my round-1 description of T-4 ("Risk-tiered approval and authorisation") specified that auto-approval events log "which rule authorised it," but I did not specify what the…” — finance_domain, round 2*

Mandatory sampling audit of the auto-approved population

> *“neither position has specified what fraction of volume is an acceptable delegation ceiling, nor what compensating control (sampling audits of auto-approved invoices? periodic re-authorisation of the delegation matrix itself?) prevents the delegated tier from silently…” — pragmatist, round 2*

Governed delegation-of-authority policy engine for the low-risk tier

> *“I accept that a governed, versioned tolerance-rule and delegation-matrix object is the right mechanism for risk-tiering” — emerging_economy, round 2*

Offline-cached delegation policy with named fallback approver

> *“I now accept that my tiered-approval design needs an explicit, auditable delegated/offline fallback authority (not a bypass, a named alternate approver with logged after-the-fact reconciliation)” — hitl_designer, round 2*

Matching kept as a discrete auditable step boundary

> *“This remains a discrete step boundary, not an inline computation inside a larger pipeline, because the boundary is the auditable point at which the in…” — council synthesis, TS-02*

Discrete subledger lock preserved under continuous reconciliation

> *“the period-end confirmed balance and the subledger lock that enforces cutoff (KB-03) must remain a hard gate — this is not something continuous reconciliation should be allowed to soften” — finance_domain, round 1*

Excluded claim

Segregation enforced at credential level — Excluded from the controls list: baseline draw 7fc2 already states "Authorisation is performed by an approver, structurally distinct from the accountant who later posts the entry." The council version is stronger (separate credentials, logs and escalation ownership, enforced architecturally) but the control is not absent from every baseline draw, so it is not claimed as council-only.

Reviewer questions

1. The delegated approval tier has no stated volume ceiling. What fraction of invoice value may be auto-authorised before the sampling audit stops being a sufficient compensating control?

2. Segregation is specified at credential level between approval and posting. Is a service account executing delegated authorisation a distinct actor for VIP-R-06 purposes, or does the gate's actor comparison need to see identity rather than role name?

3. The offline approval fallback reconciles after the fact. What is the acceptable reconciliation window before an unreconciled offline approval becomes an audit finding?

Requirements checked by the gate

ID	Rule type	Statement
VIP-R-01	step_preserved	The objective of three-way matching must survive the redesign in some form; the matching function may be retained, restructured, merged or absorbed, but not eliminated.
VIP-R-02	inputs_required	Whatever carries the matching function must still consume PO data and GRN data.
VIP-R-03	step_preserved	Invoice approval must survive in some form.
VIP-R-04	ordering_preserved	Approval must occur before payment execution.
VIP-R-05	step_preserved	GL posting must survive; invoices cannot bypass the ledger.
VIP-R-06	actor_separation	The approval function and the posting function must not be performed by the same actor.
VIP-R-07	step_preserved	AP reconciliation must survive the redesign.
VIP-R-08	step_preserved	Payment execution must survive in some form.

All 8 requirements passed in all six runs, with zero vacuous passes. Steps recorded as open to elimination by design: VIP-01, VIP-02, VIP-04.

Knowledge base as supplied to both conditions

14 entries, identical in both conditions. Every **ref** is TBC: paragraph citations are verified by hand against the standards texts and are never written from memory, so no pack quotes a paragraph number.

ID	Standard	Topic	Relevance	Bears on
KB-01	IAS 1 / LKAS 1	Accrual basis	core	PC-02, VIP-06
KB-02	IAS 1 / LKAS 1	Fair presentation and faithful representation	core	PC-05, PC-07, PC-08
KB-03	IAS 1 / LKAS 1	Reporting-period discipline	core	PC-01, VIP-08, RF-02
KB-04	IAS 8 / LKAS 8	Accounting estimates	core	PC-02
KB-05	IAS 8 / LKAS 8	Error correction	core	PC-02, PC-05, VIP-04
KB-06	IAS 8 / LKAS 8	Policy consistency	core	PC-06, RF-05
KB-07	IAS 21 / LKAS 21	FX translation	core	PC-06
KB-08	IAS 21 / LKAS 21	Exchange differences	core	PC-06, VIP-07, RF-03
KB-09	IAS 2 / LKAS 2	Inventory cost and evidence	core	VIP-03, VIP-06
KB-10	IAS 2 / LKAS 2	NRV write-downs	core	PC-02
KB-11	IFRS 9 / SLFRS 9	Payables as financial instruments	marginal	VIP-06, VIP-07
KB-12	IFRS 9 / SLFRS 9	Expected credit losses	marginal	PC-02
KB-13	IFRS 15 / SLFRS 15	Revenue recognition timing	marginal	RF-02, PC-06
KB-14	IFRS 10 / SLFRS 10	Intercompany elimination	core	PC-04, PC-06

Provenance

Runs	20260828-175146-baseline-P2P-VIP-01-fd37, 20260830-022032-baseline-P2P-VIP-01-b43c, 20260830-022117-baseline-P2P-VIP-01-7fc2, 20260829-051950-council-P2P-VIP-01-13b0, 20260830-022209-council-P2P-VIP-01-4c1d, 20260830-052308-council-P2P-VIP-01-2023
Commits	24be37cf3378e40c844b3af4fc343d9d668e4409, 561d515112c61efa253e8534e7aff998880f1ec6, 9b06cd86710bd5b068bca6bb0c76270fd29c5663
Encoding	p2p_vendor_invoice_processing.json · sha256 bda6b409a100cda43f0ecaf4edc9eefc9152d42672663cecf7db6b83c7af36cb
Knowledge base	ifrs_slfrs_kb.yaml · sha256 0b59fb3e729dc91835694ae76f04aa975d8b6eda52d05052fdf0ac7d51914d53
Model	claude-sonnet-5 (as resolved by probe)
Instrument version	v1 disposition convention (pre-01034ae)
v2 exclusion	4 v2 probe runs at commit 01034ae are excluded from this pack; runs made under different instructions are not pooled
Generated	scripts/make_review_pack.py, read-only from run artefacts

Development split. These runs informed development iteration. Reviewer comments on this pack inform development, not research data, unless separately agreed.